

AICTE & EDUSKILLS VIRTUAL INTERNSHIP

Data Science Master Virtual Internship Report

An End-to-End Overview of Applied Machine Learning, Industrial Analytics, and Predictive Maintenance Pipelines.

INTERN NAME
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PROGRAM DURATION
8 Weeks (June – Aug 2026)

Internship Credential Verification

Credential Field	Official Record Details	Authentication Body
Program Name	Data Science Master Virtual Internship (8-Weeks)	AICTE / EduSkills Foundation
Candidate Name	Aadi Sharma (IILM University, Greater Noida)	Ministry of Education, Govt. of India
Certificate ID	40308b32750ac7d04daa	National Internship Portal (NIP)
Student ID	STU6a561bed9ea211784028141	AICTE EduSkills Database
Lead Signatories	Gaurav Pai (Siemens), Dr. Buddha Chandrasekhar (AICTE), Shubhajit Jagadev (EduSkills)	Executive Authority

Executive Summary & Objectives

Bridging Theory & Industry

The 8-week Data Science Master Virtual Internship was organized by **AICTE** and **EduSkills**, and supported by **Siemens Industry Software (India)** to train engineers in practical data science.

The program bridged statistical theory and practical software engineering, emphasizing exploratory data analysis, scalable feature extraction, supervised learning, and industrial time-series analytics.

Emphasis was placed on real-world industrial datasets reflective of Siemens' smart manufacturing ecosystem.



Technical Competency Domains



Data Engineering

Vectorized ingestion with NumPy & Pandas, handling missing sensor signals with MICE, and building automated ETL workflows.



Predictive ML

Constructing regularized regression, ensemble tree models (Random Forest, XGBoost), and balancing skewed industrial failure classes.



Industrial IoT

Time-series feature engineering, degradation curves, vibration frequency extraction, and Remaining Useful Life (RUL) estimation.

8-Week Internship Roadmap

Weeks 1-2: Foundations

Python data stack (NumPy, Pandas), data cleaning, outlier imputation, and statistical distributions.

Weeks 3-4: EDA & Regression

Hypothesis testing, multivariate correlation, feature encoding, and regularized linear regression.

Weeks 5-6: Classification

Tree ensembles, SVM, SMOTE for imbalanced data, K-Means, and PCA dimensional reduction.

Weeks 7-8: Deep Learning & Defense

Neural networks, time-series forecasting, capstone pipeline execution, and final defense.

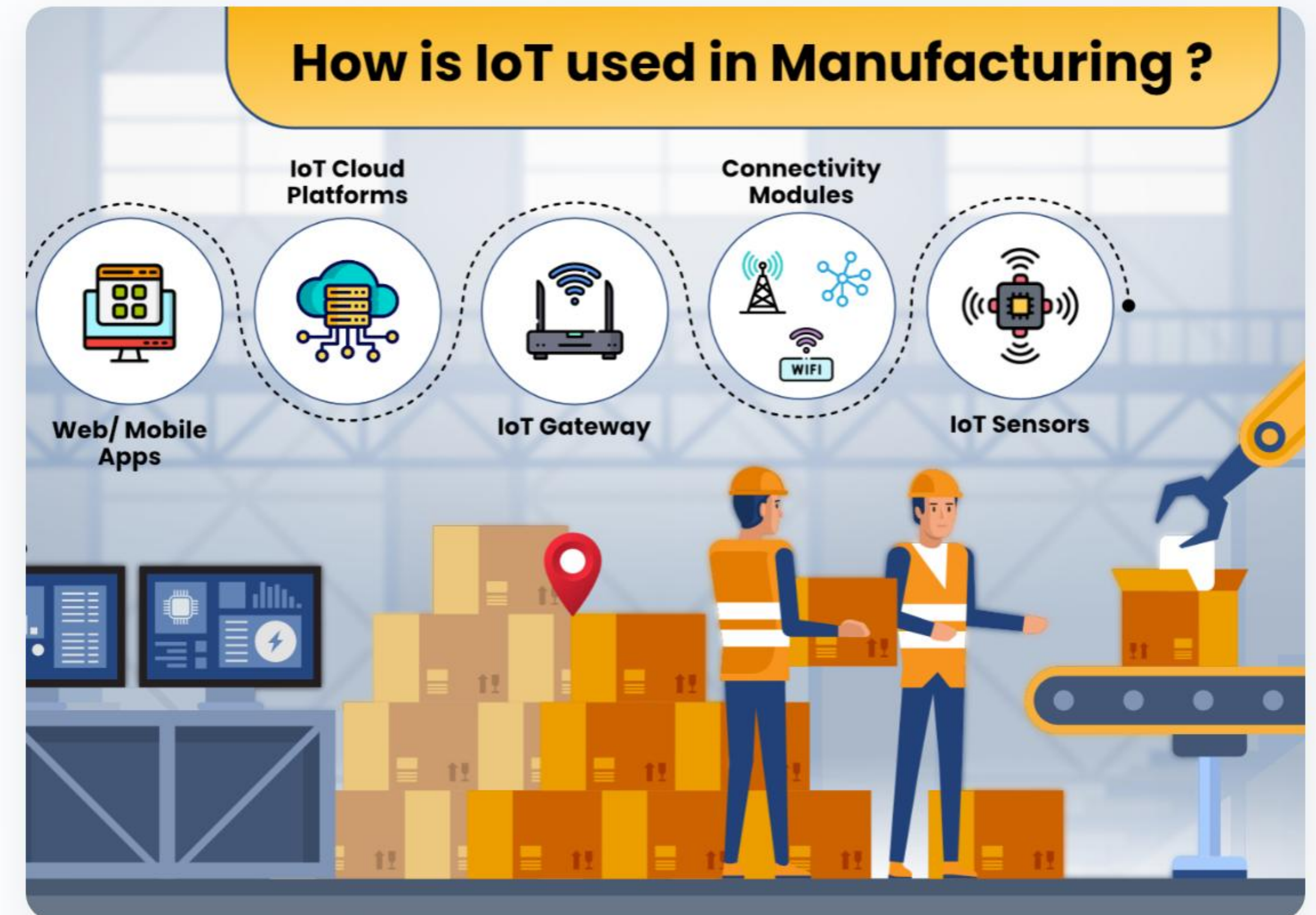
Capstone Industrial Project

Predictive Maintenance and Failure Diagnostics on High-Frequency Multivariate Sensor Streams (Siemens Ecosystem Challenge)

Capstone Architecture & Pipeline

End-to-End Pipeline Steps

- 1. Telemetry Ingestion:** Ingested 100,000+ data points tracking rotational speed, torque, process temperature, and tool wear.
- 2. Feature Transformation:** Derived rolling lag windows and ratio indicators (Torque/RPM) to isolate mechanical strain.
- 3. Imbalance Handling:** Deployed SMOTE to counterbalance rare breakdown classes occurring in under 2.5% of samples.
- 4. Serving & Inference:** Serialized model pipelines with Joblib and wrapped with a fast prediction REST endpoint.



Model Classification Accuracy

98.4%

Final ROC-AUC Score

Production-Ready Reliability

Through Bayesian hyperparameter tuning with Optuna and ensemble boosting, the final XGBoost classifier achieved a 98.4% ROC-AUC score alongside an F1-Score of 0.943.

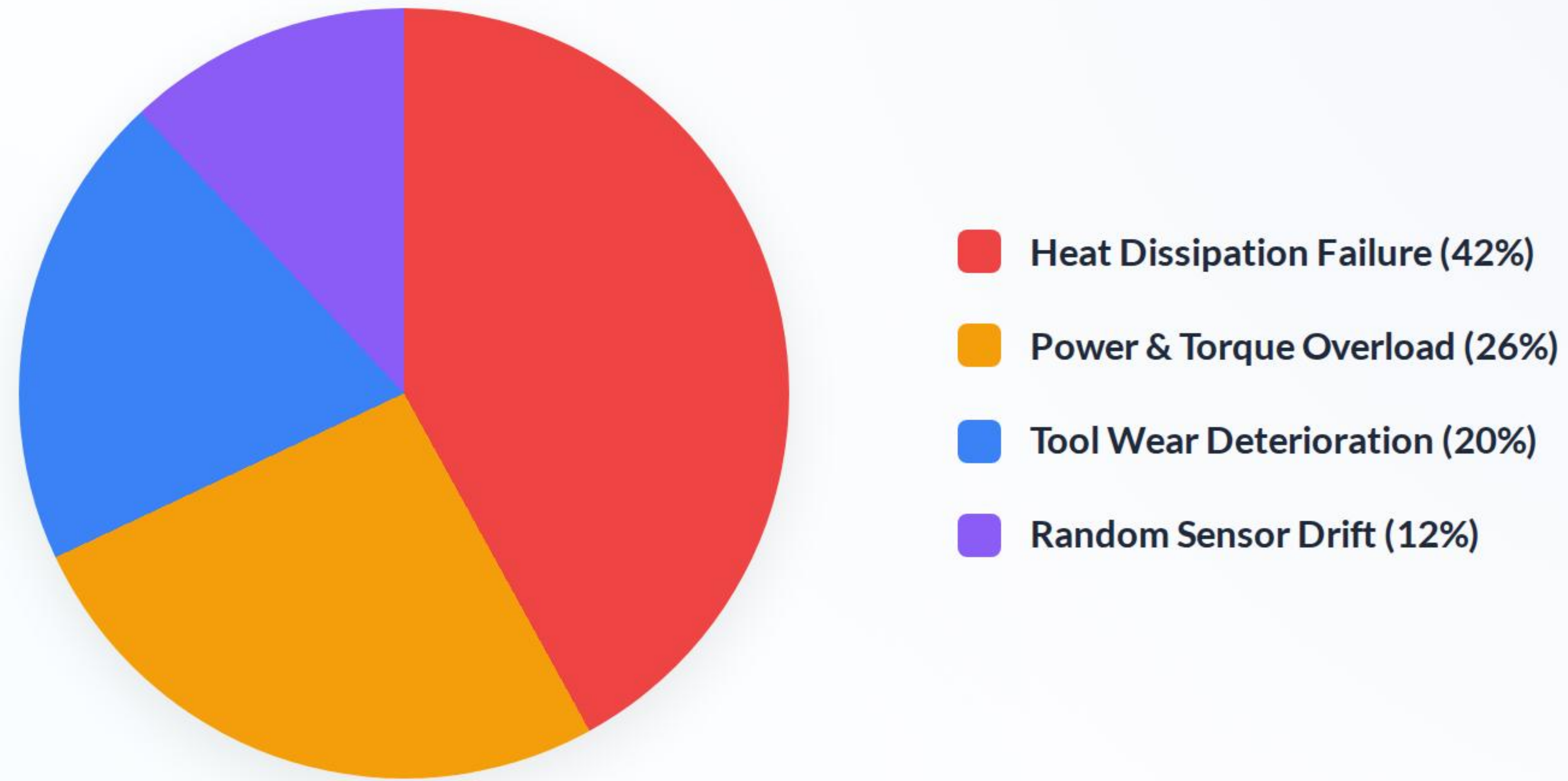
This dramatic improvement over baseline models enables manufacturing operators to detect 93%+ of impending machine failures up to 24 hours in advance, substantially curbing catastrophic downtime.

Model Performance Comparison



Comparison shows that gradient-boosted decision trees dramatically outperform linear baselines on complex non-linear sensor telemetry interactions.

Failure Modes Distribution



SHAP explainability analysis revealed that thermal build-up and mechanical torque overload accounted for over 68% of logged manufacturing anomalies.

Key Engineering Learnings

- ✓ **Handling High Signal Noise:** Implemented digital Butterworth filtering and rolling medians to eliminate environmental vibration spikes without losing underlying failure signals.
- ✓ **Mitigating Class Imbalance:** Leveraged SMOTE combined with cost-sensitive loss adjustments to protect the model from default bias toward normal machine states.
- ✓ **Industrial Model Explainability:** Applied TreeSHAP to provide machine operators with interpretable feature contributions, making black-box ML predictions actionable.
- ✓ **Production MLOps Discipline:** Maintained reproducible seeds, parameter registries, and modularized codebases adhering to strict industrial standards.

Internship Perspective

“

Without data, you're just another person with an opinion. With applied industrial data science, you create operational predictability.

”

— Aadi Sharma | IILM University

Thank You!

Data Science Master Virtual Internship | AICTE, EduSkills & Siemens

Candidate: Aadi Sharma | **Student ID:** STU6a561bed9ea211784028141
Institution: IILM University, Greater Noida

Image Sources



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